

How CPO and PSPO are related

Same constrained-RL template; different safety set, evidence, and update rule

Bottom line

PSPO is not literally a special case of the CPO algorithm. It is better viewed as a sibling constrained policy-optimisation method: CPO enforces a local expected-cost constraint, while PSPO projects PPO proposals into a formally verified, shield-defined parameter region.

A common abstraction

$$\underset{\theta}{\text{maximise}} J_R(\theta) \quad \text{subject to} \quad \theta \in \mathcal{C}_{\text{safe}}$$

The important difference is what $\mathcal{C}_{\text{safe}}$ means and how an update is kept inside it.

CPO

Expected-cost constrained trust-region step

$$\begin{aligned} \max_{\Delta\theta} g^\top \Delta\theta \\ c + b^\top \Delta\theta \leq 0 \\ \frac{1}{2} \Delta\theta^\top F \Delta\theta \leq \delta \end{aligned}$$

Here c is current expected cost minus the budget; b is the cost-surrogate gradient; and F is the KL/Fisher curvature. A dual solution, conjugate gradients, and line search produce the next policy (or a cost-recovery step).

PSPO

Verified-set constrained projected PPO

$$\begin{aligned} \mathcal{C}_H &= \{\theta : \forall s, \arg \max_a z_{\theta,a}(s) \in A_H(s)\} \\ \Theta_k^{\text{LID}} &\subseteq \mathcal{C}_H \\ \theta_{k+1} &= \text{Proj}_{\Theta_k^{\text{LID}}}(\bar{\theta}_{k+1}) \end{aligned}$$

The shield supplies admissible actions $A_H(s)$. IBP certifies a Local Invariant Domain (LID) in actor-parameter space. PPO supplies the reward-improving proposal; projection or fail-closed reversion supplies safety.

The relationship in one line



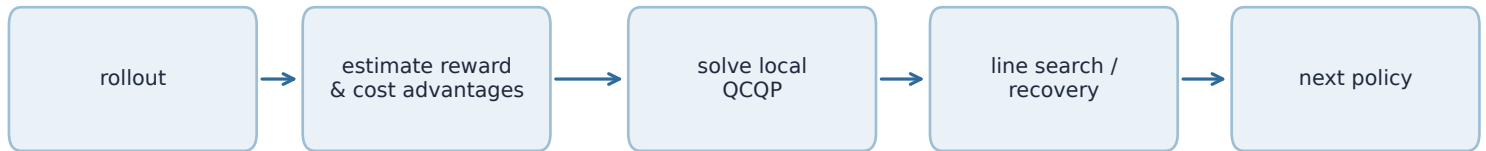
CPO changes the step direction using cost and KL; PSPO changes the feasible parameter set using the shield and LID.

Why the distinction matters

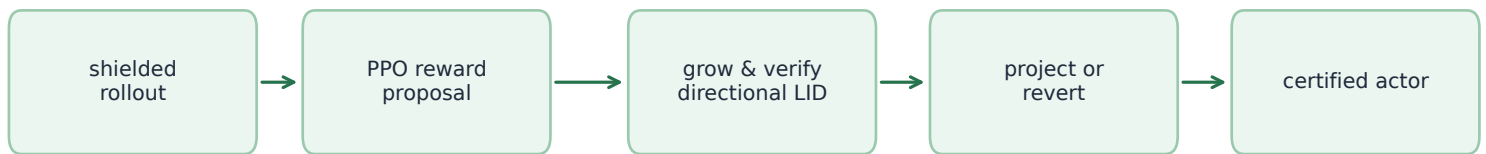
Optimisation mechanics and guarantee scope

One update

CPO



PSPO



Side-by-side

Dimension	CPO	PSPO
Safety quantity	Expected cumulative cost	Greedy shield admissibility at every certified state
Evidence	Rollout estimates and learned cost critic	Exhaustive shield data and sound interval bounds
Local geometry	KL/Fisher ellipsoid + linearised cost half-space	Certified parameter orthotope (LID)
Reward optimiser	TRPO-family natural-gradient step	PPO proposal followed by Euclidean projection
Failure handling	Backtracking or cost-recovery step	Last certified LID or fail-closed reversion
Guarantee target	Approximate satisfaction of an expected-cost budget	All-state action certificate at enforcement boundaries
Certified deployment	Usually the stochastic policy	Greedy actor; stochastic sampling needs an added certificate

Best shorthand

PSPO is projected PPO over a shield-derived, verified inner approximation of the safe-policy set. It resembles CPO at the level of constrained optimisation, but it is not obtained by choosing a special CPO cost function or trust-region radius.

When the ‘special case’ intuition is useful

A useful conceptual bridge—and its limits

Encode shield violations as cost

$$c_H(s, a) = \mathbf{1}\{a \notin A_H(s)\}, \quad J_C(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_t c_H(s_t, a_t) \right]$$

With non-negative costs and budget $d=0$, exact satisfaction forces zero shield violations on state-action pairs that receive positive occupancy. This makes the high-level feasible-policy problem look similar to PSPO.

Why that still does not make PSPO CPO

1 Occupancy versus all states

CPO constrains an expectation under the rollout distribution. PSPO certifies every enumerated state in its declared domain, including states absent from a finite rollout.

2 Approximation versus verification

CPO linearises cost and KL and estimates them from data. PSPO accepts an LID only after a hard, sound interval certificate.

3 Different step

CPO jointly chooses a reward/cost-aware trust-region direction. PSPO first computes a PPO reward proposal, then constructs a safe region and projects the proposal.

4 Different safety semantics

PSPO certifies that the greedy action is shield-admissible. Trajectory safety additionally inherits the shield's soundness; stochastic deployment requires a probability-level extension.

A unified research direction

Use CPO-style reward/cost geometry to propose efficient directions, but accept each update only inside a PSPO-style verified LID. This would combine expected-cost guidance with a hard statewise certificate; it is a hybrid, not a specialisation.

Sources: Achiam et al., Constrained Policy Optimization, ICML 2017 (arXiv:1705.10528); repository PSPO methodology (pspo_main.tex) and CPO implementation (core/safe_rl_baselines/cpo.py).